

S. Yvette Chen

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Education

University of California, Berkeley, Bachelor's Degree

2025—2029

- **GPA: 4.0/4.0** | **Dean's List** [link to transcript]
- **Coursework:** *Principles and Techniques of Data Science (DATA C100)* | *Discrete Mathematics and Probability Theory (CS 70)* | *Data Structures (CS 61B)* | *Linear Algebra and Differential Equations (MATH 54)*
- **Auditing:** *4D Seminar (LBNL Theory)* | *Computational Models of Language Use and Interaction (CS 294-258)*

Chongqing Nankai Secondary School, High School Diploma

2022—2025

- **GPA: 4.0/4.0** | **Honor Roll** [link to transcript]

Experiences

Student AI Researcher | UC Berkeley & Lawrence Berkeley National Lab (LBNL)

Jan 2026 - Present

- Collaborating within Keenan's Group to develop uncertainty-aware multi-agent architectures for *bias labeling* in Lidar-based 3D point clouds and raster data. Reduced per-tree labeling time by 98%.
- Designed and implemented a *zero-modification generalizable* decoupling framework to translate expert rules into executable decision trees, enabling the dynamic reconfiguration of the orchestrator agent without retraining, achieving Macro-F1 of 67.7% and outperforming ML baselines.

Independent ML Researcher | Jacobs Institute for Design Innovation

Sep 2025 - Present

- Designing a *NeRF*-based architecture for 3D manifold reconstruction from mesh parameters, with a *GNN* module for spinal curvature classification.
- Previously selected from 200+ applicants incl. undergraduate, Master's and Ph.D. as one of the three undergraduate teams to join the Berkeley Student Entrepreneurship Program (StEP) in Fall 2025.

Computational Science Research Intern | Southwest University

Dec 2022 - Feb 2025

- Performed data cleaning, normalization, regression modeling and population dynamics modeling to quantify the effect of E_jFUL gene edits on loquat maturation timelines.
- Work was recognized as Outstanding Project of the Year (10 out of 100+ projects received this award). Findings were published and patented (2024).

Projects, Pre-Prints & Publications

Chen, S., Saban, N., Hargreaves, C., & Wang, H. (2026). TreeAgent: a generalizable Multi-Agent framework for automated bias labeling in forestry via compiled expert rules and Vision-Language models (*ICML AI4Science*, Submitted)

- Contributed as the first author to design and implement a zero-modification generalizable framework to transpile and execute expert rules

YXiZ: NeRF-Assisted and Deep Learning-Driven Spatial Inference for At-Home Scoliosis Detection [poster]

Chen, S., & Yuan, Z. (2024). Biomarkers of Specific Brain Changes in Adolescents with Anxiety Disorder Based on MRI Multimodal Data. *Advances in Education Humanities and Social Science Research*, 12(1), 722.[link]

- Developed statistical learning pipelines using a *Linear Mixed Model in MATLAB* to perform high-dimensional, covariate-controlled analysis on a large-scale neuroimaging dataset (n>10,000).
- Applied *graph theory algorithms* to model structural covariance, using graph-based metrics (degree centrality, clustering, modularity) to characterize network dysfunction; optimized computation for multi-dataset scalability.

End-to-End Large-Scale Sentiment Classifier based on Multinomial Bayes (Finalist, National Math Modeling Competition)

- Implemented a full NLP pipeline (*tokenization, lemmatization, TF-IDF vectorization*) and trained a *multinomial Naive Bayes* classifier for sentiment classification on 100k+ TikTok comments, achieving AUC 0.85–0.89.
- Proactively enhanced model robustness by curating training sets to address complex NLP challenges (*sarcasm, slang, and multilingual text*).

Honors & Awards

The Jacobs Institute Innovation Catalysts Jacobs Institute for Design Innovation	Feb 2026
• Awarded on the project YXiZ: Deep Learning-Driven Spatial Inference for At-Home Scoliosis Detection. • The YXiZ project has previously been accepted to Berkeley Student Entrepreneurship Program (StEP) and proceeded to the semi-finalist round of the Fall 2025 cohort.	
Outstanding Project of the Year The Chuying Innovation Talent Program	Feb 2025
• Performed data cleaning, normalization, regression modeling and population dynamics modeling to quantify the effect of EjFUL gene edits on loquat maturation timelines. • Work was recognized as Outstanding Project of the Year (<i>10 out of 100+ projects receive this award</i>). Findings are published and patented (2025).	
Grand Slam International Space Settlement Design Competition (ISSDC) [Link]	Apr 2023 - Jul 2024
• Led the first-and-only Chinese delegation to achieve the Grand Slam in national, Asian, and international finals. • Space Settlement Design Competitions (SSDCs) have been held at the famous NASA's Jet Propulsion Laboratory; NASA's John F. Kennedy Space Center; NASA's Dryden Flight Research Center; NASA's Johnson Space Center; and NASA's White Sands Test Facility.	
National Finalist China National College Mathematics Modeling Competition	May 2024
Certificate of Occupational Ability of Data Analysis China Shipper's Association	May 2024
National Silver Award Canadian Chemistry Competition (CCC)	May 2024
Global Gold Award (top 5%) British Biology Olympiad (BBO)	May 2024
International Finalist (top 1%) HOSA Biochemistry Contest (HOSA)	Apr 2024

Skills

Programming: Python (primary), Java, C++ , TypeScript, Scheme

Machine Learning: scikit-learn, PyTorch, PyTorch Geometric, NLTK, OpenCV, LangGraph, HuggingFace, SpaCy

Data & Systems: Node.js, FastAPI, PostgreSQL

Scientific & Research: MATLAB, NumPy, Pandas, Matplotlib, LaTeX

Research Focus: Machine Learning, (Geometric) Deep Learning, Compositionality, Grounded Reasoning